HEALTHCARE · CASE 021

An AI that flagged chest X-rays sped up reports, not lung cancer diagnosis

In a randomized trial across five NHS trusts in England, an AI tool flagged chest X-rays it judged abnormal so they could be reported first. Reports came back sooner. The trial found no significant difference in how soon patients got a CT scan or a lung cancer diagnosis.

WHAT HAPPENED

- Between July 2023 and December 2024, five NHS trusts in England took part in a trial of chest X-rays requested by GPs. Of 97,731 X-rays taken, 93,326 were analysed.
- On days chosen at random, an AI tool, qXR from Qure.ai, sent a notice to the reporting list for each X-ray it judged abnormal, so it could be reported first. On the other days the AI still marked up the X-rays, but sent no notice. On both kinds of day, staff could still review an X-ray straight away under local procedure.
- On the AI days, X-rays were reported sooner: a median of 34.1 hours, against 47.0 hours on the other days.
- The trial found no significant difference in how soon patients got a CT scan or a lung cancer diagnosis. The median time from X-ray to CT scan was 53 days either way.
- The investigators conclude that X-ray AI should not include this kind of prioritization in this setting.

Grade B Good evidence

KEY NUMBERS

97,731
chest X-rays performed during the trial period

86,945
patients whose X-rays were analysed

34.1 h
median time to an X-ray report on AI days, against 47.0 hours on the other days

53 days
median time from X-ray to CT scan, with or without AI prioritization

WHY IT MATTERS

In a randomized trial across five NHS trusts, AI prioritization made chest X-ray reports come back sooner, but the trial found no significant difference in how soon patients got a CT scan or a lung cancer diagnosis. The library's reading: a faster step in a care pathway is not the same as a faster result for patients, and this trial measured both.

STILL UNKNOWN

- What AI prioritization costs. The paper reports no cost figures for this trial; the investigators say a health economics evaluation will be published separately.
- The paper does not state whether Qure.ai received any part of the trial's grant or was paid for the use of qXR.
- How many of the X-rays the AI flagged were in fact reported straight away. The paper does not report it.

Read the full record, with every source and its quotes: https://danielwipert.github.io/ai.case.study/cases/2026/aai-2026-021-lungimpact-ai-chest-xray-prioritization-rct/

Everything above except "Why it matters" is the case's checked summary text; "Why it matters" is the library's reading, checked against the same record. Checked 1 oct 2026. Made 12 Oct 2026.